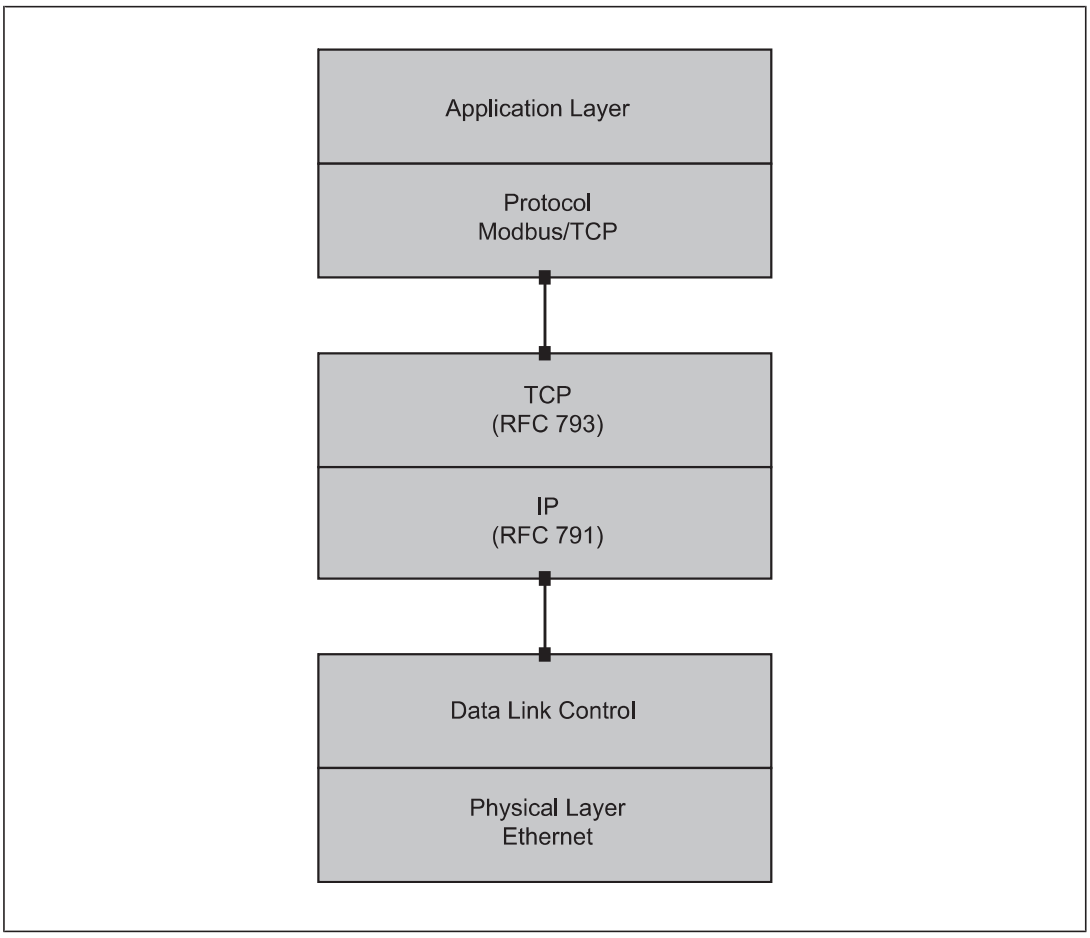
6 Modbus/TCP

6.1 System requirements

- PNOZmulti Configurator: From Version 9.0.0
- PNOZ m B0 + PNOZ m ES ETH
- PNOZ m B1

6.2 Modbus/TCP - Basics

Modbus/TCP is an open fieldbus standard published by the user organisation MODBUS-IDA.



Modbus/TCP is a protocol based on Industrial Ethernet (TCP/IP over Ethernet). It is one of the protocols with Client/Server communication. Data is transferred via a request/response mechanism using function codes (FC).

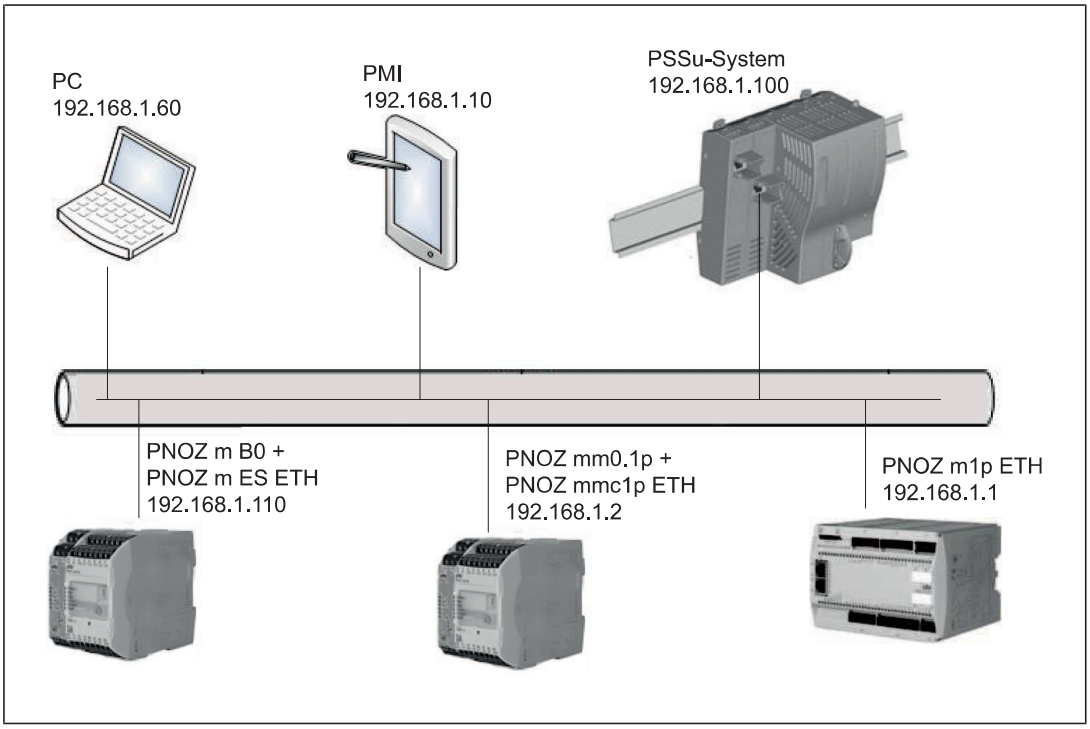
Modbus/TCP is connection-oriented, i.e. before the payload is transferred via Modbus/TCP, a connection must first be established between two Modbus/TCP interfaces. The initiator of the connection is called the Client. The communications partner with which the Client establishes the connection is called the Server. Whether a connection on a device assumes the role of Client or Server is defined when a connection is configured. As a result, the Server/Client role only applies for that specific connection.

6.3 Modbus/TCP with PNOZmulti 2

On PNOZmulti 2, an interface for communication via Modbus/TCP is provided via the communication module PNOZ m ES ETH or the integrated interface of the PNOZ m B1.

A PNOZ m ES ETH can manage a max. of 8 Modbus/TCP connections. The PNOZmulti 2 is always the connection Server. The connection Clients may be various devices, e.g. PC (PNOZmulti Configurator), controller, display unit. They can access the configurable control system PNOZmulti 2 simultaneously.

The virtual I/Os plus all the information that is requested during fieldbus communication are contained in data areas. The data is accessed directly. It's no longer necessary to switch via table/segment.



The configurations required for Modbus/TCP are fully pre-configured in the operating system of the PNOZmulti 2. Only the virtual inputs and outputs need to be activated in the PNOZmulti Configurator (see PNOZmulti Configurator Online Help).

On a configurable control system PNOZmulti 2, port number "502" is fixed as the default for data exchange via a Modbus/TCP connection. It is not displayed in PNOZmulti Configurator and cannot be modified.

6.4 Data areas

6.4.1 Overview

A configurable control system PNOZmulti 2 supports the following Modbus/TCP data areas:

Data area	Modbus syntax	Example
Coils (Bit) 0x00000 … 0x65535 [read/write]	0x[xxxxx]	0x00031 (virtual input i31)
Discrete Inputs (Bit) 1x00000 … 1x65535 [read only]	1x[xxxxx]	1x08193 (virtual output o1)
Input Register (Word/16 Bits) 3x00000 … 3x65535 [read only]	3x[xxxxx]	3x00002 (virtual inputs 32 ... 47)
Holding Register (Word/16 Bits) 4x00000 … 4x65535 [read/write]	4x[xxxxx]	4x04108 (project name, 1st character)



INFORMATION

Addressing for PNOZmulti systems starts at “0”. On devices from other manufacturers, addressing can start at “1”.
Please refer to the operating manual provided by the relevant manufacturer.

Data can be accessed via various Modbus/TCP data areas.

The tables below show the relationship between Modbus/TCP data areas and the content of the data areas.

6.4.2 Function codes

The following function codes (FC) are available for communication with the PNOZmulti 2 via Modbus/TCP:

Function code	Function	
FC 01	Read Coils	The connection Client reads bit data from the connection Server, data length ≥ 1 Bit, content: Input/output data (data received from 0x)
FC 02	Read Discrete Input	The connection Client reads bit data from the connection Server, data length ≥ 1 Bit, content: Input/output data (data received from 1x)
FC 03	Read Holding Register	The connection Client reads word data from the connection Server, data length ≥ 1 Word, content: Diagnostic word (data received from 4x)

Function code	Function	
FC 04	Read Input Register	The connection Client reads word data from the connection Server, data length ≥ 1 Word, content: Diagnostic word (data received from 3x)
FC 05	Write Single Coil	The connection Client writes to one bit datum in the connection Server, data length = 1 Bit, content: Input data (send data to 0x)
FC 06	Write Single Register	The connection Client writes to one word datum in the connection Server, data length = 1 Word, content: Input data (send data to 4x)
FC 15	Write Multiple Coils	The connection Client writes to multiple bit data in the connection Server, data length ≥ 1 Bit, content: Input data (send data to 0x)
FC 16	Write Multiple Registers	The connection Client writes to multiple word data in the connection Server, data length ≥ 1 Word, content: Input data (send data to 4x)
FC 23	Read/Write Multiple Registers	The connection Client reads and writes multiple word data within a telegram (receive data from 3x and send data to 4x)

Error codes on Modbus/TCP

Code	Name	Description
01	Invalid function	The function code in the enquiry is not supported.
02	Invalid data address	The data address received in the enquiry is outside the memory range.
03	Invalid data	Invalid data requested.

6.4.3 Data transfer limits

The following table contains information on the data length per telegram that is supported:

Data transfer		Data length per telegram
Read data (Bit)	FC 01 (Read Coils)	1 … 2000
	FC 02 (Read Discrete Inputs)	
Read data (Bit)	FC 05 (Write Single Coil)	1 Bit
	FC 15 (Write Multiple Coils)	1 … 1968

Data transfer		Data length per telegram
Read data (Word)	FC 03 (Read Holding Registers)	1 … 125
	FC 04 (Read Input Register)	
Write data (Word)	FC 06 (Write Single Register)	1 Word
	FC 16 (Write Multiple Registers)	1 … 123 Words
Read and write data (Word)	FC 23 (Read/Write Multiple Registers)	Read 1 … 125 Words Write 1 … 121 Words



INFORMATION

There may be some restrictions in data length, depending on the device that is used. Please refer to the information stated in the operating manual of the device you are using.

6.4.4 Input and output data, watchdog

The table below describes the Modbus/TCP data areas that contain the current state of the virtual inputs of the PNOZmulti 2. These are the virtual inputs that can be set by the user.

Relevant areas for the data are defined in each Modbus/TCP data area (Coils (0x), Discrete Inputs (1x), Input Register (3x), Holding Register (4x)). Read/write access will depend on the Modbus/TCP data area.

Register (3x, 4x)	Coil/ Discrete Input (0x, 1x)	Content	High Byte	Low Byte
0	15... 0	State of virtual inputs 0...15	i15…i8	i7…i0
1	31... 16	State of virtual inputs 16...31	i31…i24	i23…i16
2	47...32	State of virtual inputs 32...47	i47…i40	i39…i32
3	63...48	State of virtual inputs 48...63	i63…i56	i55…i48
4	79... 64	State of virtual inputs 64...79	i79…i72	i71…i64
5	95...80	State of virtual inputs 80...95	i95…i88	i87…i80
6	111...96	State of virtual inputs 96...111	i111…i104	i103…i96
7	127...112	State of virtual inputs 112...127	i127…i120	i119…i112

The table below describes the Modbus/TCP data areas that contain the states of the virtual outputs of the PNOZmulti 2.

Relevant areas for the data are defined in the Modbus/TCP data areas Discrete Inputs (1x) and Input Register (3x). Read access is available to these data areas.

Register (3x)	Discrete Input (1x)	Content	High Byte	Low Byte
512	8207...8192	State of virtual outputs 0...15	o15…o8	o7…o0
513	8223...8208	State of virtual outputs 16...31	o31…o24	o23…o16
514	8239...8224	State of virtual outputs 32...47	o47…o40	o39…o32
515	8255...8240	State of virtual outputs 48...63	o63…o56	o55…o48
516	8271...8256	State of virtual outputs 64...79	o79…o72	o71…o64
517	8287...8272	State of virtual outputs 80...95	o95…o88	o87…o80
518	8303...8288	State of virtual outputs 96...111	o111…o104	o103…o96
519	8319...8304	State of virtual outputs 112...127	o127…o120	o119…o112

A watchdog can be activated in Control Register 255.
If no input bits are set by a Modbus/TCP subscriber within the preset time, the PNOZmulti 2 will set the input bits to "0".

The table below describes the Modbus/TCP data areas for the watchdog.

A relevant area for the watchdog is defined in each Modbus/TCP data area (Coils (0x), Discrete Inputs (1x), Input Register (3x), Holding Register (4x)). Read/write access will depend on the Modbus/TCP data area.

Register (3x, 4x)	Coil/ Discrete Input (0x, 1x)	Content	High Byte	Low Byte
255	4095...4080	Control Register	See table below	

High Byte	WD-Trigger	Error Message	Reserved	Reserved	Reserved	W-Timer Bit 2	W-Timer Bit 1	W-Timer Bit 0
Low Byte	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

Bit 15 "Watchdog Trigger": The watchdog can be triggered by setting Bit 15 constantly to "1", or by a Client writing in the input area of the 128 inputs. The state of the bit when reading is undefined. A 1 or a 0 can be read.

Bit 14 "Error Message": If this bit is set, an error stack entry is generated when the watchdog is triggered.

Bit 10 … 8 "WD Timer": If the set time for the watchdog is set, then Bit 15 must be set, or become set, at the same time.

Watchdog Timer Bit 2	Watchdog Timer Bit 1	Watchdog Timer Bit 0	Watchdog Time
0	0	0	Timer deactivated
0	0	1	100 ms
0	1	0	200 ms
0	1	1	500 ms
1	0	0	1 s
1	0	1	3 s
1	1	0	5 s
1	1	1	10 s



INFORMATION

To check whether the watchdog has been triggered, set a virtual input permanently to "1".
If the corresponding input in the PNOZmulti 2 is "0", then the watchdog has been triggered.

6.4.5 Allocation table of the virtual inputs and outputs

Addressing the virtual inputs (coils) in Registers 0 to 7.

	Virtual inputs															
	15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
0	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
2	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
3	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48
4	79	78	77	76	75	74	73	72	71	70	69	68	67	66	65	64
5	95	94	93	92	91	90	89	88	87	86	85	84	83	82	81	80
6	111	110	109	108	107	106	105	104	103	102	101	100	99	98	97	96
7	127	126	125	124	123	122	121	120	119	118	117	116	115	114	113	112

Allocation of virtual inputs i0, i1 ... i127 to the Bits in Registers 0 to 7

Addressing the virtual outputs (discrete inputs) in Registers 512 to 519.

	Virtual outputs															
	15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
512	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
513	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
514	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
515	63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48
516	79	78	77	76	75	74	73	72	71	70	69	68	67	66	65	64
517	95	94	93	92	91	90	89	88	87	86	85	84	83	82	81	80
518	111	110	109	108	107	106	105	104	103	102	101	100	99	98	97	96
519	127	126	125	124	123	122	121	120	119	118	117	116	115	114	113	112

Allocation of virtual outputs o0, o1 ... o127 to the Bits in Registers 512 to 519.

6.4.6 Service data

The service data for Modbus/TCP is divided into the following data areas:

- Process data - Base unit and expansion modules
- Process data - Fieldbus module and communication module
- Diagnostic words
- Enable elements
- Project data
- Device data

The content of the version and device-dependent service data and addressing are described in the chapter entitled Service data [📖 86].

6.4.7 LEDs

The table below describes the Modbus/TCP data areas that contain the LED statuses.

Relevant areas for the data are defined in the Modbus/TCP data areas Discrete Inputs (1x) and Input Register (3x). Read access is available to these data areas.

Register (3x)	Discrete Input (1x)	Contents	High Byte	Low Byte
511	8176	Status of LEDs	Reserved	PNOZmulti 2 LEDs

The LED status of the base unit PNOZ m B0 can be requested directly as follows

- Bit 0 = 1: LED OFAULT is lit or flashes
- Bit 1 = 1: LED IFAULT is lit or flashes
- Bit 2 = 1: LED FAULT is lit or flashes
- Bit 3 = 1: LED DIAG is lit or flashes
- Bit 4 = 1: LED RUN is lit
- Bit 5-7: Reserved

The LED status of the base unit PNOZ m B1 can be requested directly as follows

- Bit 0 = 1: LED OFAULT is lit or flashes
- Bit 1 = 1: LED IFAULT is lit or flashes
- Bit 2 = 1: LED FAULT is lit or flashes
- Bit 3 = 1: LED DIAG is lit or flashes
- Bit 4 = 1: LED RUN FS is lit
- Bit 5: Reserved
- Bit 6 = 1: LED RUN ST is lit
- Bit 7: Reserved

6.4.8 Updating the data areas

Data is updated with varying priority.

The table below shows the typical update cycles for the various data.

Content	Typ. update cycle
Virtual inputs/outputs	20 ms
Project and device data	Once during initialisation
State of the inputs/outputs from the base unit and expansion modules	320 ms
LED status	1000 ms
Number of elements that can store a state	Once during initialisation
Element enable	320 ms
Diagnostic words	1000 ms
Current state of the virtual inputs	1000 ms



INFORMATION

The update time may increase if there are additional TCP/IP connections (e.g. PNOZmulti Configurator, PMI, control system) on the PG port (Port 9000).